

# API SECURITY RISK ANALYSIS REPORT

Future Interns Cyber Security Internship (2026)

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Submitted By

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## 1. API SELECTED FOR ANALYSIS

API Used: <https://jsonplaceholder.typicode.com>

This is a public REST API used for testing and learning purposes.

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## 2. PURPOSE OF ANALYSIS

The objective of this task is to analyze a public API and identify common security risks such as authentication issues, data exposure, and access control weaknesses.

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## 3. API ENDPOINTS ANALYZED

The following endpoints were reviewed:

- /users
  - /posts
  - /comments
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## 4. SECURITY RISKS IDENTIFIED

### 1. Open Access (No Authentication)

The API does not require login credentials or API keys.

Impact:

- Anyone can access API data
- No user verification system

Severity: HIGH

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### 2. Excessive Data Exposure

API returns full dataset without filtering sensitive fields.

Impact:

- Risk of sensitive data exposure in real systems
- Unnecessary data disclosure

**Severity: MEDIUM**

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### **3. No Rate Limiting**

**There are no restrictions on number of requests.**

**Impact:**

- API can be abused with repeated requests
- Risk of service overload

**Severity: MEDIUM**

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### **4. Lack of Authorization Controls**

**No role-based access or permission checks exist.**

**Impact:**

- All users can access all data
- Privacy and security risks in real applications

**Severity: HIGH**

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## **5. RISK CLASSIFICATION TABLE**

| <b>Risk</b>              | <b>Severity</b> | <b>Business Impact</b>          |
|--------------------------|-----------------|---------------------------------|
| <b>No Authentication</b> | <b>High</b>     | <b>Unauthorized data access</b> |
| <b>Data Exposure</b>     | <b>Medium</b>   | <b>Information leakage</b>      |
| <b>No Rate Limiting</b>  | <b>Medium</b>   | <b>API abuse / overload</b>     |
| <b>No Authorization</b>  | <b>High</b>     | <b>Privacy violation</b>        |

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## **6. REMEDIATION RECOMMENDATIONS**

- Implement authentication (JWT / OAuth2)
- Use role-based access control (RBAC)
- Enable API rate limiting
- Return only required data fields

- **Use HTTPS for secure communication**
  - **Validate all API inputs and outputs**
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## **7. CONCLUSION**

**API security is essential in modern applications. Weak APIs can lead to data breaches, unauthorized access, and system abuse. Proper authentication, authorization, and rate limiting must be implemented to secure APIs effectively.**

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**END OF REPORT**